UNIVERSAL CATENARITY OF A LOCALLY NOETHERIAN SCHEME IS LOCAL, AND PASSES TO SCHEMES LOCALLY OF FINITE TYPE OVER IT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 3118-3132 and proof span 3134-3161 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the statement's conditions by numbers (1) to (4); those references are the author's and are preserved.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- An open immersion is locally of finite type and a composite of morphisms locally of finite type is locally of finite type, both cited from the same chapter. Cited at proof lines 3351-3352. Shows that open subschemes and schemes locally of finite type over a universally catenary base are again universally catenary.
- The algebra lemma that over a universally catenary ring every finite type algebra is catenary. Cited at proof lines 3360-3361. Converts the hypothesis on one affine chart into universal catenarity of its coordinate ring.
- The same-chapter characterization of locally finite type morphisms on affine charts, and the properties chapter lemma that catenarity is local on a scheme. Cited at proof lines 3372-3373 and 3376-3377. Make the coordinate rings of the chosen affine cover of X finite type algebras over the base rings, and then assemble catenarity from the pieces.

Source reference. Original lemma 02I2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3285–3285. Same-chapter reference used in the retained author text at source line(s) 3147. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01T3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2360–2360. Same-chapter reference used in the retained author text at source line(s) 2889, 2937, 3138, 3329, 9763, 9913. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2362–2362. Same-chapter reference used in the retained author text at source line(s) 2980, 3135, 3326, 9944. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3316–3316. Same-chapter reference used in the retained author text at source line(s) 3160. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a locally Noetherian scheme. The following are equivalent*

1. *S is universally catenary,*
2. *there exists an open covering of S all of whose members are universally catenary schemes,*
3. *for every affine open $\mathrm{Spec}(R) = U \subset S$ the ring R is universally catenary, and*
4. *there exists an affine open covering $S = \bigcup U_i$ such that each U_i is the spectrum of a universally catenary ring.*

Moreover, in this case any scheme locally of finite type over S is universally catenary as well.

Proof. By Lemma 01T5 an open immersion is locally of finite type. A composition of morphisms locally of finite type is locally of finite type (Lemma 01T3). Thus it is clear that if S is universally catenary then any open and any scheme locally of finite type over S is universally catenary as well. This proves the final statement of the lemma and that (1) implies (2).

If $\mathrm{Spec}(R)$ is a universally catenary scheme, then every scheme $\mathrm{Spec}(A)$ with A a finite type R-algebra is catenary. Hence all these rings A are catenary by Algebra, Lemma 02I2. Thus R is universally catenary. Combined with the remarks above we conclude that (1) implies (3), and (2) implies (4). Of course (3) implies (4) trivially.

To finish the proof we show that (4) implies (1). Assume (4) and let $X \to S$ be a morphism locally of finite type. We can find an affine open covering $X = \bigcup V_j$ such that each $V_j \to S$ maps into one of the U_i. By Lemma 01T2 the induced ring map

$\mathcal{O}(U_i) \to \mathcal{O}(V_j)$ is of finite type. Hence $\mathcal{O}(V_j)$ is catenary. Hence X is catenary by Properties, Lemma 02IX. □